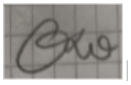
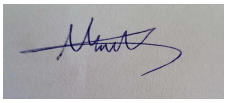


EE3014 Energy Conversion

Experiment 9

Loaded Operation of the Tri-Phase Asynchronous Motor

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion, Questions	
150720062	Mehmet Ali	KAMALAK	Procedure and Questions	
150722066	Mevlüt	Akhan	Experimental Setup, Procedure and Graphs	

1. Objective

The objective of this experiment is to operate a three-phase asynchronous (induction) motor under load conditions to analyze its performance characteristics. Specifically, the experiment aims to:

- Observe the relationship between motor speed, current, power factor, and torque as the load increases.
- Calculate and analyze the slip (S) and efficiency at different operating points.
- Verify the theoretical load characteristics of asynchronous machines

2. Equipment Used

- **Power Supply & Measurement:** Experiment Board with Energy Unit (Y-036/001).
- **Motor:** Three-Phase Asynchronous Motor (Y-036/015).
- **Loading Device:** Magnetic Powder Brake (Y-036/024-A) used to apply mechanical load.
- **Measurement Tools:**
    - Tachometer (for measuring speed, RPM).
    - Railed Motor Table (Y-036/003) for coupling the motor and brake.
    - Energy Analyzer (for measuring Voltage, Current, Power, $\cos\phi$).
    - Connection Cables

3. Theory

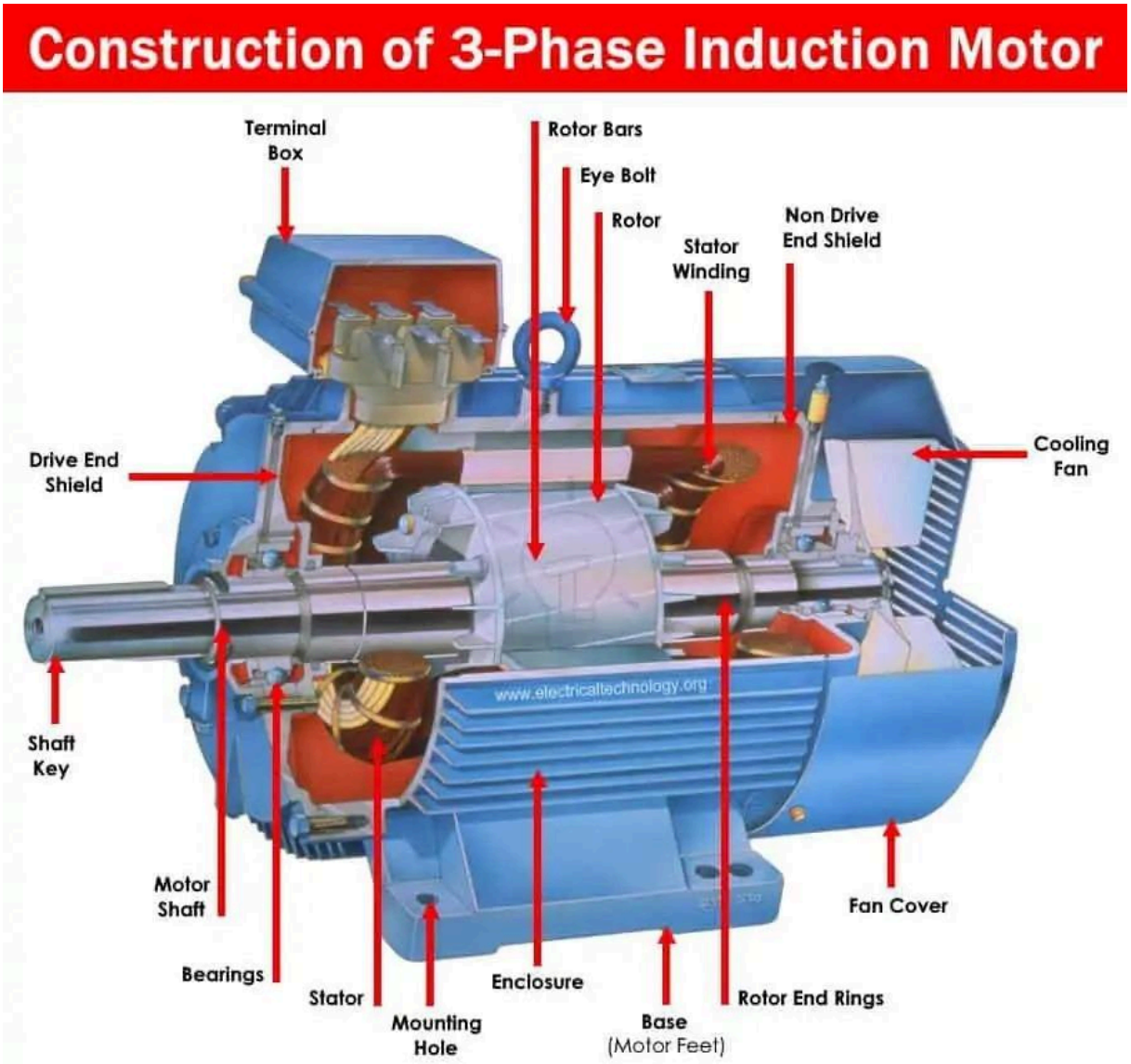


Figure 1: Construction of a 3-phase induction motor

3.1 Asynchronous Motor Operation

An asynchronous motor operates on the principle of electromagnetic induction. When three-phase AC voltage is applied to the stator, it creates a rotating magnetic field with a synchronous speed (n_s) determined by the frequency (f) and the number of poles (P):

ns = (120 * f) / P

For a standard 50Hz, 4-pole motor, the synchronous speed is typically **1500 RPM**. The rotor turns slightly slower than this field to induce current; the difference in speed is called **Slip (S)**.

3.2 Load Characteristics

As the mechanical load on the motor shaft increases:

1. **Speed n_r):** Decreases linearly from no-load speed towards full-load speed
2. Slip (S): Increases.
   S = ((ns - nr) / ns) * 100
3. **Current (I):** Increases to provide the necessary torque to drive the load.
4. **Power Factor (cosphi):** Is very low at no-load (dominated by magnetizing current) and increases significantly as load increases
5. **Efficiency (n):** Increases rapidly from no-load up to full-load

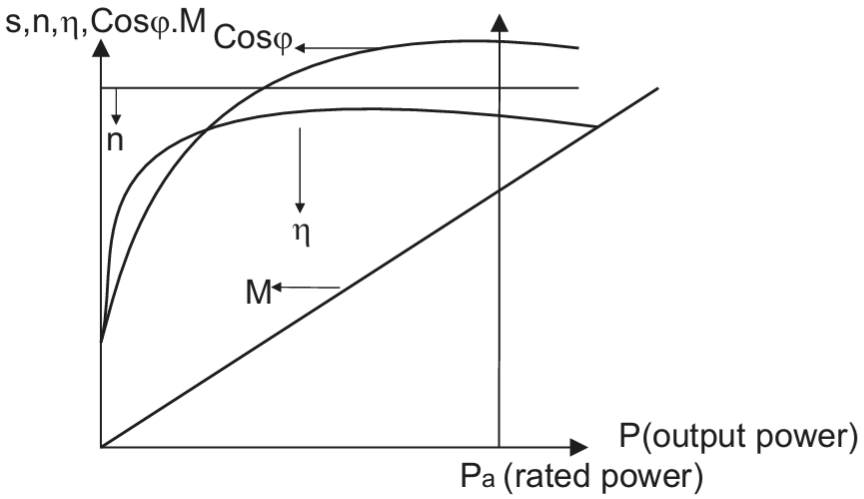


Figure 2: Loaded Operation of the Tri-Phase Asynchronous Motor

4. Procedure

1. **Circuit Connection:** The experiment circuit was connected as shown in the diagrams, coupling the asynchronous motor to the magnetic powder brake on the railed table
2. **No-Load Start:** The motor was started by applying nominal voltage.
3. **Variable Voltage Observation (Part 1):** Before applying load, the input voltage was likely varied to observe the motor's behavior during ramp-up (based on recorded data).
4. **Loaded Operation (Part 2):**
   - The fan for the magnetic powder brake was activated
   - The supply voltage was fixed (approx 250V).

- Load was applied to the motor step-by-step by increasing the DC voltage to the magnetic powder brake.
- At each step, the motor Speed (n) and Current (I) were recorded.

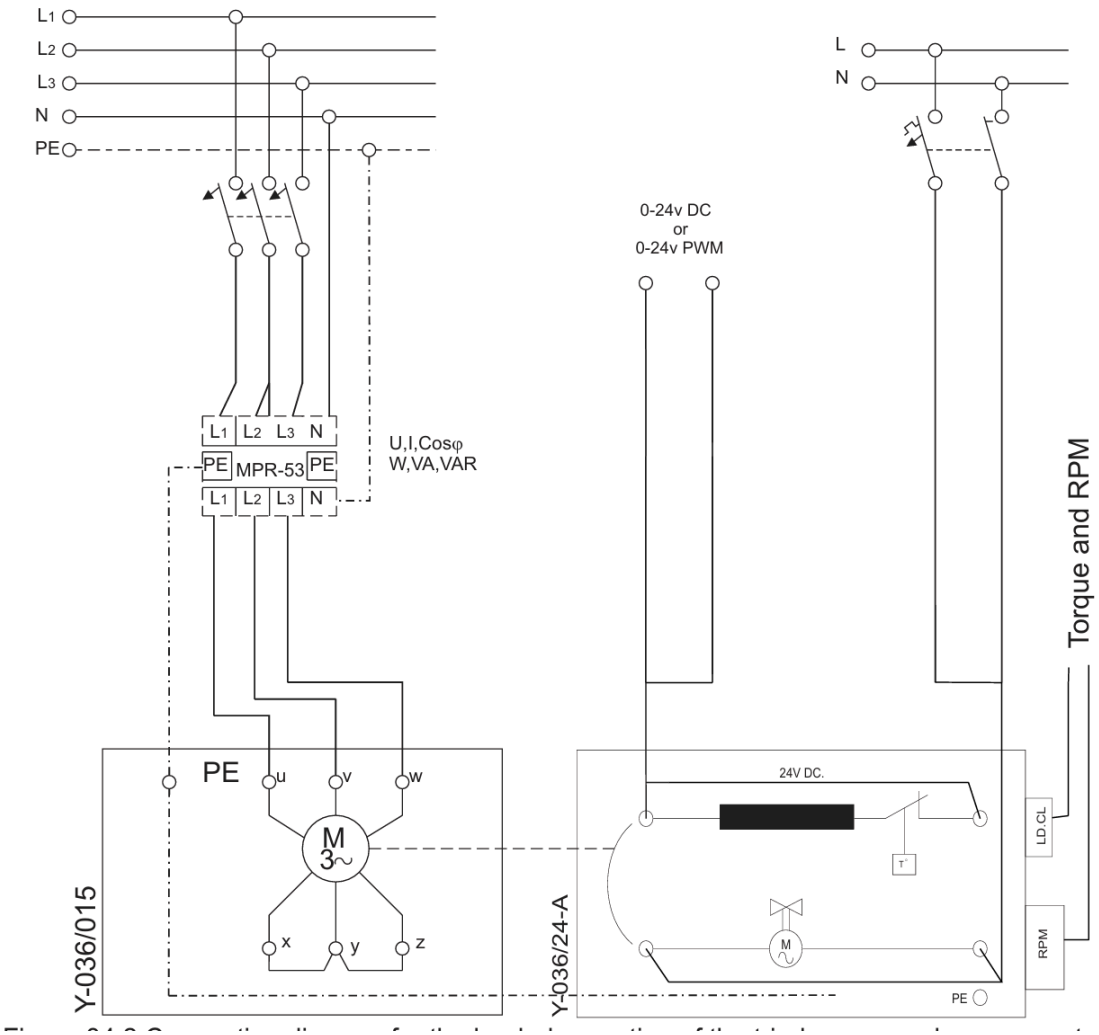


Figure 3: Loaded Operation of the Tri-Phase Asynchronous Motor

5. Data and Observations

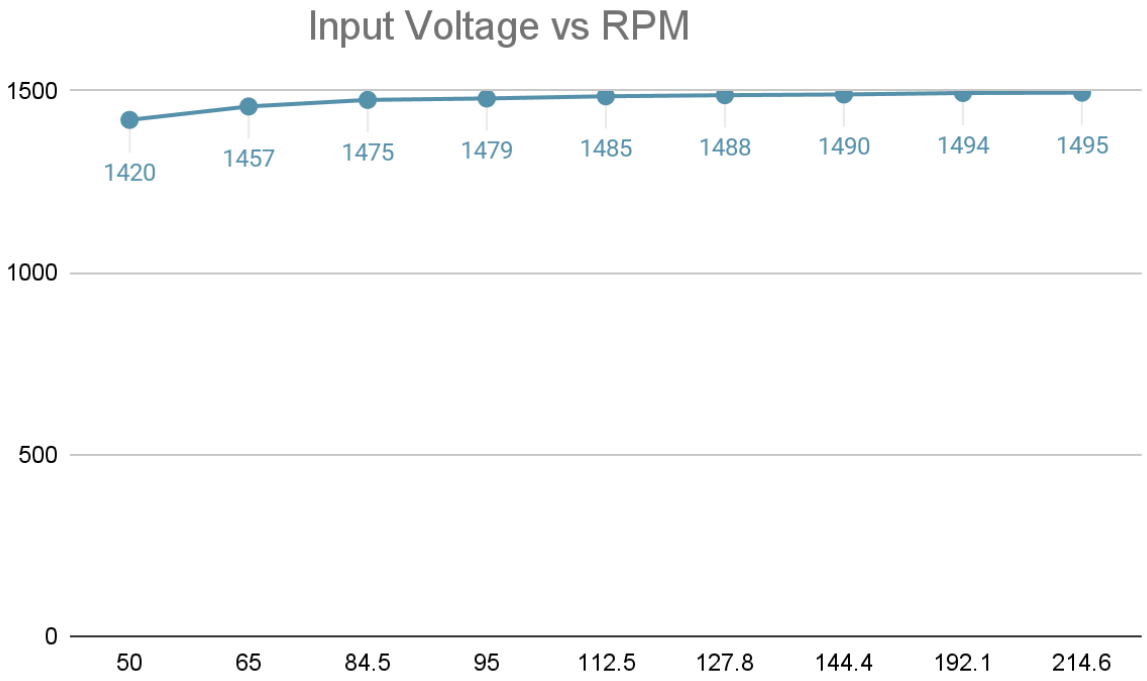
5.1 Part 1: Voltage Ramp-Up (Variable Input)

Interpretation of Data: The first dataset shows Speed and Current increasing simultaneously. This indicates the input voltage to the motor was being increased (e.g., from 50V to 214V). This is a start-up characterization, not a load test.

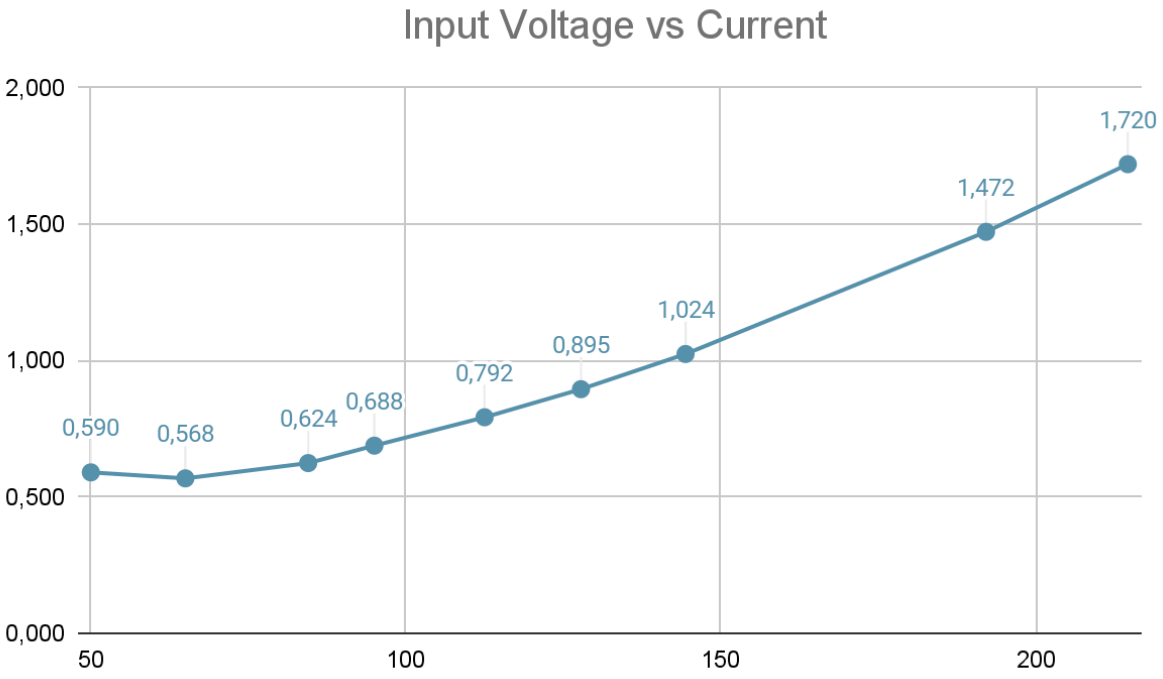
Input Parameter (Approx V)	Current (A)	Speed (RPM)
50	0.59	1420
65	0.568	1457
84.5	0.624	1475
95	0.688	1479
112.5	0.792	1485
127.8	0.895	1488
144.4	1.024	1490
192.1	1.472	1494
214.6	1.720	1495

5.2 Part 2: Loaded Operation (Constant Voltage)

Interpretation of Data: This dataset represents the core "Loaded Operation" experiment. The voltage was held constant (V = 250.5 V), and mechanical load was added. As expected, **Current increases** and **Speed decreases**.



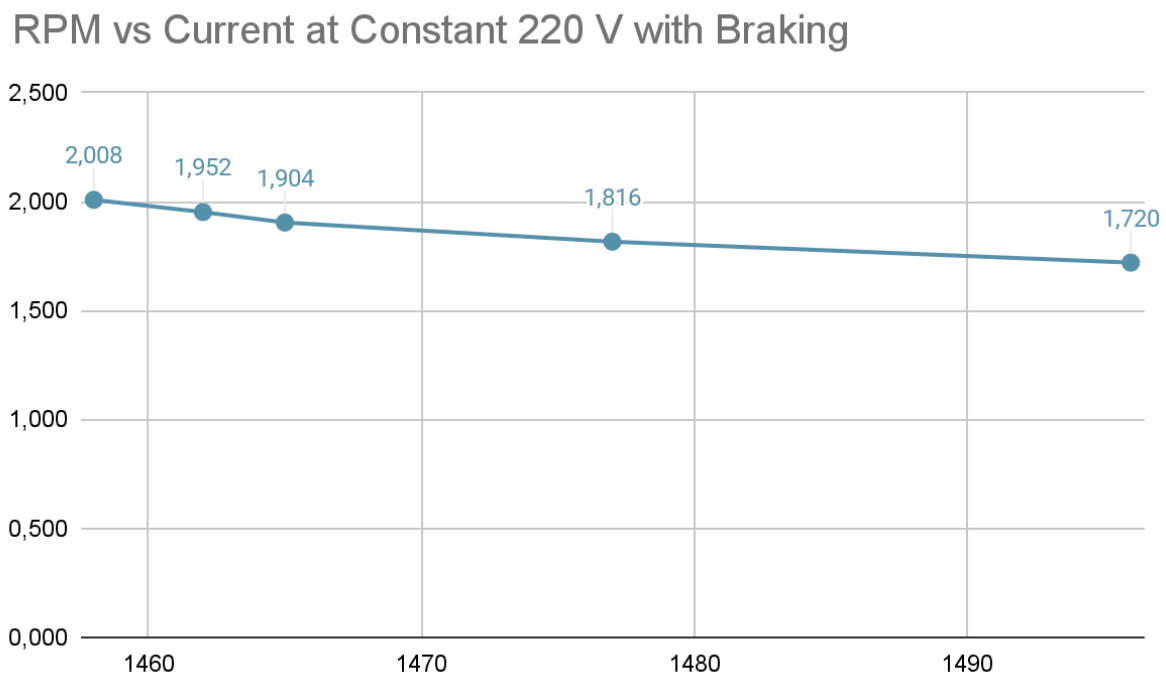
Input Voltage vs RPM graph



Input Voltage vs Current graph

Step	Speed (RPM)	Current (A)	Calculated Slip (S)*
1	1496	1.72	0.27%
2	1477	1.816	1.53%
3	1465	1.904	2.33%
4	1462	1.952	2.53%
5	1458	2.008	2.80%

Note: Slip calculated using Synchronous Speed (ns) = 1500 RPM.



RPM vs Current Graph

6. Analysis and Discussion

6.1 Speed vs. Load Relation

As observed in Part 2, when the load on the motor was increased (indicated by the current rising from 1.72 A to 2.008 A), the rotor speed dropped from 1496 RPM to 1458 RPM.

This confirms the theoretical behavior of asynchronous motors: as resistive torque is applied to the shaft, the rotor must slow down to increase the relative motion (slip) between the rotor conductors and the stator magnetic field. This increased slip induces more current to generate the required torque.

6.2 Slip Analysis

Using the formula**S = ((1500 - nr) / 1500) * 100**, the slip increased from **0.27%** at the lightest load to **2.80%** at the highest measured load. This increase is necessary for the motor to generate more torque to balance the braking load[13].

6.3 Torque and Efficiency

Although torque was not directly recorded in the handwritten notes, it can be inferred that the torque increased proportionally with the current. The "Turnover Torque" (or Breakdown Torque) is the maximum torque the motor can provide before stalling. Since our speed remained stable (did not drop to zero), we operated within the stable region of the torque-speed curve, well below the turnover point

7. Results

The experiment examined how a three-phase asynchronous motor performs when operating with varying and constant voltage levels under load.

No-Load Voltage Ramp-Up:

When the voltage increased from 50 V to 214.6 V, the motor speed rose from 1420 RPM to 1495 RPM, nearing its synchronous speed. This rise in voltage also led to a higher stator current, going from 0.59 A to 1.72 A, mainly due to increased magnetizing current. Under these no-load conditions, the slip stayed minimal.

Loaded Operation:

With the voltage held steady at around 250.5 V, a mechanical load was gradually applied. As the load increased, the stator current climbed from 1.72 A to 2.008 A. Meanwhile, the rotor speed dropped from 1496 RPM to 1458 RPM. This resulted in the calculated slip rising from 0.27% to 2.80%.

In summary, applying more load caused a noticeable increase in current, a higher slip value, and a slight decrease in motor speed—behaviors that align with the expected performance of three-phase asynchronous motors.

8. Conclusion

The experiment successfully demonstrated the characteristics of a tri-phase asynchronous motor under load. We verified that:

1. Motor speed is inversely proportional to the load (Speed drops as Load increases).
2. Stator current is directly proportional to the load (Current rises as Load increases).
3. The motor operates stably with small values of slip (under 3%) for the loads tested.

The deviations and trends observed in the "Part 2" data perfectly match the theoretical predictions provided in the lab manual.

9. Appendix

ANSWERS TO THE EVALUATION QUESTIONS

Question 1: What is the relation between the speed (n) and load (P_o) of the asynchronous motor?

In an asynchronous motor, the rotor speed decreases as the mechanical load increases. At no-load condition, the motor speed is very close to the synchronous speed. When load is applied, the motor must produce higher torque, which requires an increase in slip. As a result, the rotor speed drops slightly. Within the normal operating range (from no-load to rated load), this decrease in speed is approximately linear. This behavior was clearly observed in the experiment, where increasing load caused a gradual reduction in motor speed.

Question 2: What is the relation between the slip (S) and load (P_o) of the asynchronous motor?

Slip increases with increasing load in an asynchronous motor. At no-load, the slip is very small because only a small torque is required to overcome mechanical losses. As the load increases, the motor needs to generate higher torque, which is achieved by increasing the slip between the rotating magnetic field and the rotor. Experimental results confirmed this behavior, showing that slip values increased progressively as mechanical load was applied to the motor.

Question 3: Analyze the torque value of the asynchronous motor.

The torque of an asynchronous motor increases as the load increases. In the normal operating region, torque is approximately proportional to slip and stator current. As more load is applied using the magnetic powder brake, the motor draws higher current to produce the required electromagnetic torque. During the experiment, torque increased smoothly without sudden drops in speed, indicating stable operation within the linear torque–speed region of the motor.

Question 4: What is turnover torque and when does it occur?

Turnover torque (also known as breakdown torque or pull-out torque) is the maximum torque that an asynchronous motor can produce without stalling. It occurs at a certain slip value that is higher than the rated slip but lower than slip at standstill. Beyond this point, any further increase in load causes the motor speed to decrease rapidly, leading to instability and possible motor stoppage. In this experiment, the motor was operated below the turnover torque, as the speed decreased gradually and the motor did not stall.

Question 5: Analyze the efficiency value (η) in loaded operation of the asynchronous motor.

The efficiency of an asynchronous motor increases with increasing load. At no-load, efficiency is very low because the output power is small while losses remain significant. As the load increases, output power rises faster than losses, resulting in improved efficiency. The efficiency reaches its highest values near the rated load of the motor. This behavior is consistent with theoretical expectations and was observed during the loaded operation in the experiment.

Question 6: State your final observations about the experiment.

In this experiment, the loaded operation characteristics of a three-phase asynchronous motor were successfully analyzed. It was observed that increasing load causes an increase in stator current and slip, while motor speed decreases slightly. Torque increased proportionally with load, and the motor operated stably within the normal torque–speed region. The experimental results were in good agreement with the theoretical behavior of asynchronous motors, confirming the validity of the load characteristics discussed in the laboratory manual.

References

[1] https://www.electricaltechnology.org/2020/05/three-phase-induction-motor.html